# Lexer Benchmarks

## Stress Test

### System

- Architecture: ARM Apple M3 Air
- Compiler Flags: -O3 -flto
- Cache: Warm
- CPU Usage: 1 core fully utilized (98.5%)

### Input

- File: compiler_grade_test.lang
- Lines: 36,023,263
- Size: ~1.06 GB

### Methodology

- Single-threaded lexer
- 5 runs per input
- Time measured via clock_gettime(CLOCK_MONOTONIC)

### Code Used

```
#include "lexer.h"
#include <time.h>

double get_time() {
   struct timespec t;
   clock_gettime(CLOCK_MONOTONIC, &t);
   return t.tv_sec + t.tv_nsec * 1e-9;
}

int main() {
   char *name = "./compiler_grade_test.lang";

   FILE *f = fopen(name, "rb");
   fseek(f, 0, SEEK_END);
   double size = ftell(f) / (1024.0 * 1024.0);
   rewind(f);
   fclose(f);
```

```c
    int cycles = 5;
    double time = 0;
    double throughput = 0;
    long tokens = 0;

    for (int i = 0; i < cycles; i++) {
        Scanner *s = init_scanner(name);
        double start = get_time();

        Token cur;
        do {
            cur = next_token(s);
            tokens++;
        } while (cur.type != TOKEN_EOF);

        double end = get_time();

        double cycle_time = end - start;
        double cycle_throughput = size / cycle_time;

        time += cycle_time;
        throughput += cycle_throughput;
    }

    double average_time = time / cycles;
    double average_throughput = throughput / cycles;

    printf("Average time to scan: %.4f seconds\n", average_time);
    printf("Average throughput: %.2f MB/s\n", average_throughput);
    printf("Tokens produced (per cycle): %ld\n", (long)(tokens / cycles));

    return 0;
}
```

## Results

```
-----------------Benchmark results-----------------
Average lexing time: 6.6315 seconds
Average throughput: 153.06 MB/s
Tokens produced (per cycle): 168,011,034
---------------------------------------------------
```

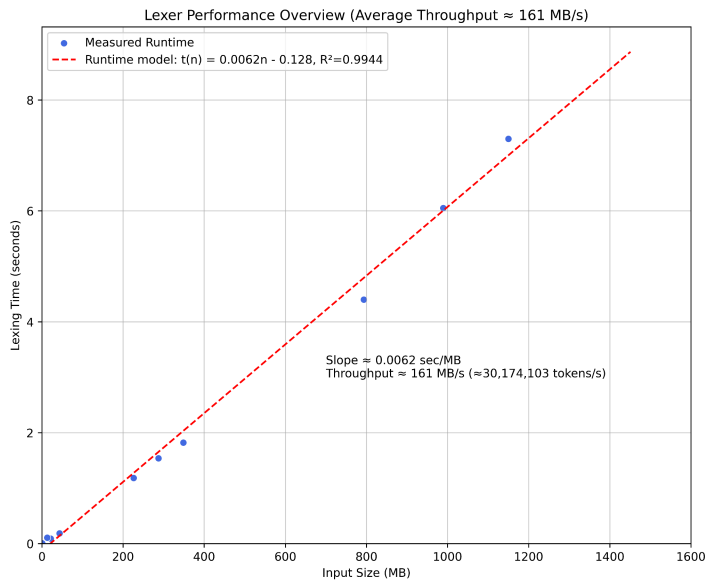
# Further Testing

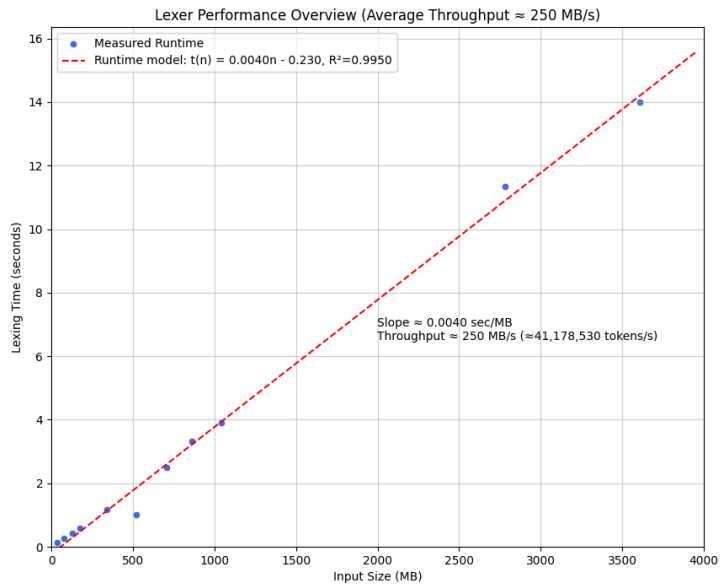
## ALL TESTS CONDUCTED IN SIMILAR PROJECT SETUPS

### Conditions:

- Compiler Flags: -O3 -flto

### Results

| Device / Architecture | Results |
|---|---|
| **Apple M3 Air (ARM)** |  <br>*Token throughput computed using lexer's true emitted token count during benchmarking.* |

| | |
|---|---|
| **Apple M1 Pro (ARM)** |  <br>*Token throughput computed using lexer's true emitted token count during benchmarking.* |

## Notes:

- Lexer time complexity: O($n$)
    - Shown by the linear regression results ($R^2$ > 0.99)
- Throughput is consistent across file size
- Negative intercept in regression line is explained by noise